Visual Recognition using Deep Learning HW1

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1 Introduction

This homework studies supervised image classification for visual recognition. The goal is to build a robust classifier that can recognize object categories under large variations in scale, viewpoint, background clutter, and intra-class appearance. Instead of relying on a plain backbone with standard global average pooling, our method combines a pretrained ResNetRS-101 backbone with multi-scale feature extraction, CBAM attention, GeM pooling, strong data augmentation, and lightweight test-time augmentation (TTA).

The core idea is straightforward: lower-level and higher-level features capture different visual cues, so combining them can improve recognition; CBAM helps the model focus on informative channels and spatial regions; GeM pooling preserves more discriminative responses than ordinary average pooling; and strong augmentation regularizes the model so that it generalizes better on unseen validation data. In short, this work aims to turn a strong pretrained backbone into a more robust classification pipeline rather than using the backbone almost unchanged.

2 Method

2.1 Data Preprocessing

We used separate preprocessing strategies for training and evaluation. During training, the input images were augmented by random resized cropping to 480×480, random horizontal flipping, RandAugment, normalization with ImageNet mean and standard deviation, and Random Erasing. These augmentations were designed to improve robustness against changes in object scale, pose, and background clutter.

For validation and testing, we adopted a deterministic pipeline consisting of resizing, center cropping, and normalization. This kept the evaluation protocol stable and reduced randomness during model comparison. In addition, we explicitly fixed the class-to-index mapping by sorting class names numerically, which prevented label-order inconsistency between the training and validation sets.

2.2 Model Architecture

Our classifier is built on top of a pretrained ResNetRS-101 backbone [1]. We extracted feature maps from three intermediate stages using `out_indices=(2,3,4)` in order to capture low-, mid-, and high-level visual cues. The input image resolution was set to 480×480 and the backbone was initialized with ImageNet pretrained weights.

To refine the extracted features, we applied the Convolutional Block Attention Module (CBAM) [2]. CBAM sequentially applies channel attention and spatial attention so that the network can emphasize informative responses while suppressing distracting background patterns. In our implementation, the channel-attention reduction ratio was set to 16 and the spatial-attention convolution kernel size was set to 7.

After attention refinement, each feature map was aggregated using Generalized Mean (GeM) pooling [3]. Compared with ordinary global average pooling, GeM provides a more flexible aggregation mechanism and better preserves strong discriminative activations. The GeM parameter was initialized with $p = 3.0$ and $\varepsilon = 10^{-6}$.

The pooled multi-scale features were concatenated and passed to a classification head composed of Batch Normalization, Dropout, a fully connected projection to a 256-dimensional hidden representation, ReLU activation, another Batch Normalization layer, a second Dropout layer, and a final linear classifier. The main dropout rate was 0.3 and the second dropout rate was 0.15.

2.3 Training Strategy

We trained the network using cross-entropy loss with label smoothing. To further regularize learning, we incorporated both MixUp [4] and CutMix [5]. During each iteration, a batch could be processed by MixUp, CutMix, or left unchanged. These mixed-sample augmentation strategies encourage smoother decision boundaries and reduce overconfident fitting.

Optimization was performed with AdamW. We used differential learning rates: 1×10^{-4} for the pretrained backbone and 5×10^{-4} for newly added modules, including CBAM, GeM, and the classification head. The weight decay was set to 1×10^{-2}, the minimum learning rate in cosine annealing was 1×10^{-6}, the label smoothing factor was 0.1, and the early-stopping patience was 20. The batch size was 16 and the training lasted for 30 epochs. Gradient clipping was additionally applied to avoid unstable updates.

2.4 Inference and Evaluation

During inference, we applied lightweight TTA by averaging logits from multiple transformed views of the same input, including the original center-cropped image, a horizontally flipped version, and a slightly larger resized view. Evaluation was mainly based on validation accuracy and cross-entropy loss. We also inspected the training and validation curves to determine whether the optimization process was stable and whether the model generalized well.

3 Results and Discussion

The proposed model showed stable convergence throughout training. The validation accuracy increased consistently over epochs, while the validation loss decreased gradually, suggesting that the model learned increasingly discriminative representations without obvious late-stage collapse. The results also indicate that the full pipeline, including multi-scale features, CBAM, GeM pooling, and strong augmentation, is trainable in a stable manner.

One detail is worth emphasizing: the training accuracy should not be interpreted too literally in this setting. Because the training pipeline includes strong augmentation and mixed-sample supervision, such as RandAugment [6], Random Erasing, MixUp, and CutMix, the conventional top-1 training accuracy is intentionally conservative and is not directly comparable to the validation accuracy. Therefore, the validation curves provide a more reliable estimate of the model's practical performance.

Overall, the current evidence suggests that the proposed design is effective for this homework setting. The validation curves rise smoothly and the loss trends remain controlled, which supports the claim that the architecture and training strategy generalize better than a naive single-scale fine-tuning pipeline.

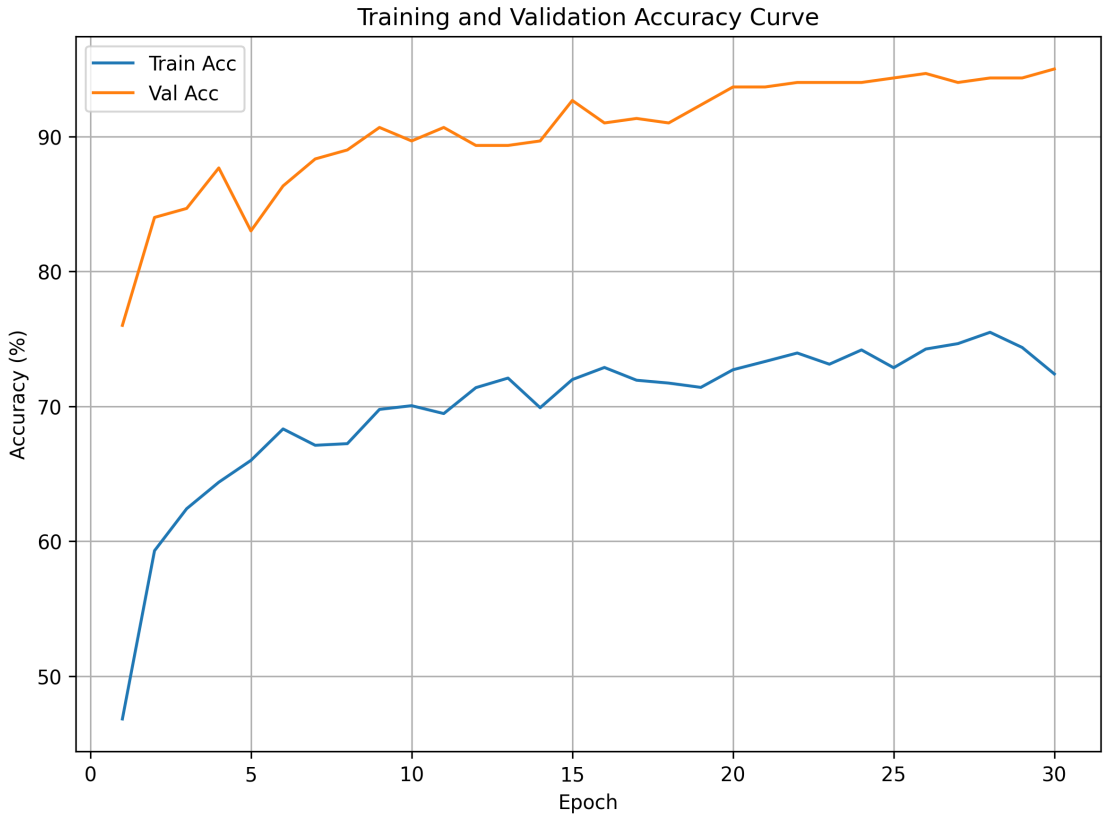

Figure 1: Training and validation curves of the proposed model. Validation trends are more reliable than training accuracy in this setting because the training phase uses strong augmentation and mixed-sample supervision.

3.1 Additional Experiments

To strengthen the empirical analysis, we conducted an ablation study by removing the CBAM modules while keeping the rest of the training pipeline unchanged.

Hypothesis. CBAM is designed to enhance feature representations through channel and spatial attention. Removing it may reduce the model's ability to focus on discriminative regions, but the overall impact may be limited due to the strong backbone, multi-scale feature fusion, and data augmentation.

Experiment setup. We removed all CBAM modules and replaced them with identity mappings. All other components, including the backbone, GeM pooling, classification head, optimizer, scheduler, and augmentation strategies, were kept unchanged to ensure a fair comparison.

Results. The model without CBAM achieved a validation accuracy of 94.67%, while the full model with CBAM reached approximately 95%. This indicates a consistent but modest improvement of about 0.3% brought by the attention mechanism. The validation loss also showed slightly better convergence when CBAM was included.

Implication. The results suggest that CBAM provides a small but consistent gain by refining feature representations. However, the relatively limited improvement indicates that most of the model's performance already comes from the backbone, multi-scale feature fusion, and strong augmentation strategy. Therefore, CBAM acts as a refinement module rather than a dominant factor in performance.

3.2 Current extension ideas

Based on the current observations, there are several plausible directions for further improvement. First, an attention-guided local crop branch could be added so that the network predicts from both the global image and the most informative local patch. This may help when the target object occupies only a small region. Second, class-balanced learning or focal loss may reduce bias toward easy categories if the class distribution or difficulty is uneven. Third, ensembling different backbones may improve robustness by combining complementary visual biases. Finally, a metric-learning objective such as ArcFace-style supervision or contrastive learning could encourage stronger inter-class separation for fine-grained categories.

References

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